

A Framework for Extracting Heart Rate Variability Features from Earbud-PPG for Stress Detection

Bhanu Teja Gullapalli
University of California San Diego
San Diego, USA
bgullapalli@ucsd.edu

Viswam Nathan
Digital Health Lab
Samsung Research America
Mountain View, USA
viswam.nathan@samsung.com

Md Mahbubur Rahman
Digital Health Lab
Samsung Research America
Mountain View, USA
m.rahman2@samsung.com

Jilong Kuang
Digital Health Lab
Samsung Research America
Mountain View, USA
jilong.kuang@samsung.com

Jun Alex Gao
Digital Health Lab
Samsung Research America
Mountain View, USA
alex.gao@samsung.com

Abstract—Unmanaged stress can lead to serious physiological and mental health issues, making it important to monitor stress continuously using wearable sensors. In this work, we investigate the use of photoplethysmography (PPG) sensors in consumer-grade earbud devices to classify periods of stress. We propose a framework that initially employs deep learning to predict heart rate variability (HRV) features from the relatively noisy earbud PPG signal. Subsequently, we transfer this knowledge and use these features for the downstream task of stress classification. The proposed framework outperforms an existing state-of-the-art library on HRV feature extraction, resulting in at least 5% improvement in accuracy, sensitivity, and specificity for stress detection.

Index Terms—Photoplethysmography (PPG), heart rate variability (HRV), transfer learning, temporal convolutional networks (TCN)

I. INTRODUCTION

Stress is the body's physical and emotional reaction in response to internal or external stressors. It is ubiquitous, with at least 1 in 2 Americans stressed during the day and around 1 million workers missing work due to stress according to the American Institute of Stress and American Psychological Association [1]. These numbers are higher than ever after the COVID-19 pandemic, inflation, and global uncertainty. Beyond immediate symptoms such as sleep deprivation or headaches, unchecked stress can lead to severe cardiovascular or neurological illness. Many individuals can be unaware of the timing and extent of their stress episodes, making it challenging to utilize interventions such as meditation and mindfulness to alleviate stress. The efficiency of any stress management solution would be greatly improved by continuous and real-time detection of stress episodes over the course of each day.

Digital health solutions, particularly wearable devices, offer a convenient and low-cost avenue for detecting stress episodes. Several studies have shown that the sensor modalities in wearable devices are capable of automatically detecting stress. In earlier works, built-in smartphone accelerometer sensors

were used to characterize the subject's behavior which is subsequently used to classify stress levels [2], [3], though this can be prone to false positives and may not be discerning enough to specifically detect stress. In one previous work, a smartphone and smartwatch were used to capture electrodermal activity (EDA) to passively detect stress in field with 75% accuracy [4]. Another study employed a custom-made wearable chest band for passive ECG sensing and detected stress with a 0.71 F1-score [5]. There has also been an approach that combined a commodity chestband to capture ECG and a commodity wristband called Empatica for EDA sensor to detect stress with 0.94 F1-score [6]. Stress detection using head-worn devices is relatively less common. Another recent work demonstrated a head-neck device capable of simultaneously capturing ECG and EEG, which can detect 'stressed' states with 75.8% accuracy [7]. While these results show promise for the use of wearable devices in daily stress detection, a common limitation, aside from smartwatches and smartphones, is that these devices are expensive, invasive for everyday use, and typically serve only a single purpose. Earbuds, as a wearable, have been gaining traction and increasingly adopted. From a signal processing perspective, they offer certain advantages over smartwatches. These include lower susceptibility to motion, a relatively fixed orientation, and their more consistent wear pattern among users. Despite these advantages, earbuds are more prone to noise interference than the more stable but less conveniently wearable options like chest bands or finger sensors. Therefore, earbuds could greatly benefit from the application of more sophisticated signal processing and machine learning approaches. The main contribution of our work was designing a transfer-learning based framework for heart rate variability (HRV) feature extraction from noisy earbud PPG signal. We also show the stress detection performance using our proposed framework on human subjects with reference sensors undergoing validated stressor tasks.

II. DATA COLLECTION AND CLEANING

A. Study Protocol

This was an experimental study approved by the Institutional Review Board of the University of California, San Francisco. The aim of this study was to detect stress from the PPG signals captured by earbud sensors. Eighteen subjects were recruited for this study, and they were all healthy individuals from the 19-22 age group to ensure that the stress responses induced are minimally affected by other confounders and comorbidities. Throughout the study, subjects wore earbuds that captured PPG at 25Hz, accelerometer at 50Hz, and Core body temperature (CBT) at 1Hz. The reference ground truth PPG from subjects was captured using a Biopac device placed on the finger that sampled PPG at 1000Hz. For our current work, we will only focus on the PPG signal.

The study involved each subject completing a set of validated tasks that induced stress, interspersed with tasks that help them relax. The study started with a baseline-recording for 5 minutes where the subject sat idle. The subject then performed a multiple object tracking task for 3 minutes (cognitive stressor), followed by 3 minutes of progressive muscle relaxation. The subject then underwent a “Speech” task; a research assistant, playing the role of an interviewer, asked the subject to speak on a challenging and personal topic (social stressor). The subject was then given 3 minute time to prepare for the speech, after which they entered a room consisting of evaluators. The research assistant purposefully delayed the start of the speech to increase the stress. After a 2-minute delay, the subject was asked to leave without completing the task. Subsequently, the subject listened to relaxing music for 4 minutes, followed by exciting music. Finally, the subject was asked to rest and recover. Between each task, the subject was given at least a 2-minute break to offset the effect of the previous task and prepare for the next task. In Figure 1, we show the timeline of the tasks and how the heart rate varies during these tasks for a sample subject.

B. PPG signal-denoising

The raw PPG signals captured from the Earbud and Biopac were first filtered by a third-order Butterworth bandpass filter with cut-off frequencies set to 0.6-4 Hz. The filtered PPG signal was then sliced into 30-second sliding windows with a 20-second overlap for the prediction of HRV parameters and stress detection. Even after passing the PPG signal through a bandpass filter, the signal still contained motion artifacts, shifts in baseline wandering, and other high-frequency noises. To address this, we estimated the quality of the PPG signal using the PQR signal quality index [8]. In this method, we calculated P/Q/R indexes representing the signal quality for each window of PPG signal. The “P” index indicated the degree of high-frequency noise in the signal, calculated using the change in the number of extremum points after bandpass filtering. The “Q” index indicated the amount of baseline wandering, calculated using the signal’s maximum and minimum amplitude. The “R” index represented the frequency of motion artifacts

by calculating the dispersion of peak and valley points. Using these three indexes, the method calculated rSQI (relative signal quality index) to compare the quality of the PPG signal against a clean reference signal. In our case, we used the signal collected from the baseline-recording as a reference signal. The PPG-windowed signals whose $rSQI \geq 0$, indicating the signal was comparable or better in quality than the reference signal, were considered for subsequent analysis. The PPG-windowed signals with $rSQI < 0$ are removed. Finally, we standardized the PPG signals of each subject to remove any subject-dependent information. Due to poor contact on one or more of the earbuds or Biopac finger sensor resulting in signals too noisy to be usable, we excluded PPG data from four subjects.

III. TRANSFER LEARNING BASED FRAMEWORK FOR EXTRACTING HRV FEATURES

To extract HRV based features from the denoised earbud PPG signal, we design a framework based on transfer learning and using Temporal convolution network (TCN) [9]. The framework consists of two steps: In the first step we pre-train a TCN model on Biopac PPG to detect position of peaks. For the latter step, we use this pre-trained TCN model for the downstream task of predicting different HRV parameters. We explain these steps in detail below.

A. Pre-training using Biopac PPG

The growing availability of ubiquitous and unobtrusive wearable devices has made the collection of continuous sensor data more convenient and accessible than ever. However, labeling this data requires significant effort and time. As a result, the amount of labeled sensor data available is minimal in comparison to the unlabeled data that can be collected. We first pre-train a model with the reference PPG signals captured from the Biopac finger sensor to detect peak positions of this input signal. We chose this task because all the HRV-related features extracted from the PPG signal are derived from the position of peaks. The primary motivation for pre-training a model on the PPG data is to increase data efficiency given the relative scarcity of available earbud PPG data, and to acquaint the model with the PPG signal. We use HeartPy [10], [11], an open-source python library for extracting ground truth peak positions. The idea of initiating models on hand-designed tasks before deploying them for different downstream purposes is not uncommon. In computer vision, models are often pre-trained with image segmentation tasks [12], [13] and in natural language processing to generate language models like BERT [14] and GPT-3 [15], models are pre-trained with tasks of predicting of next word in a sentence. We used TCN as our choice of model, while there are several choices for modeling continuous time-series data TCNs are a type of convolution network that convolve over a time-domain by combining properties of Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN). They overcome the problem of exploding and vanishing gradients by using causal dilated convolution

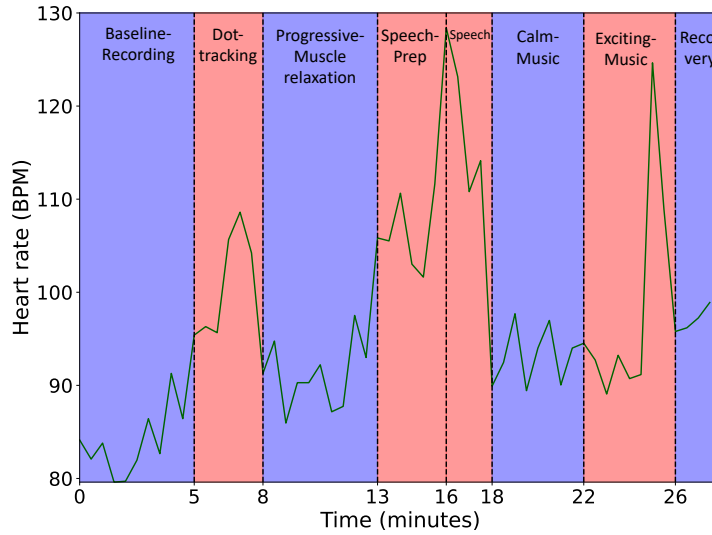


Fig. 1: Plot showing the duration and sequence of tasks that are in the study. Tasks that induce stress are marked in red, while the blue ones represent non-stressful tasks. We removed breaks in-between tasks for the sake of visualization. We also show how heart rate varies over the study for a sample individual subject.

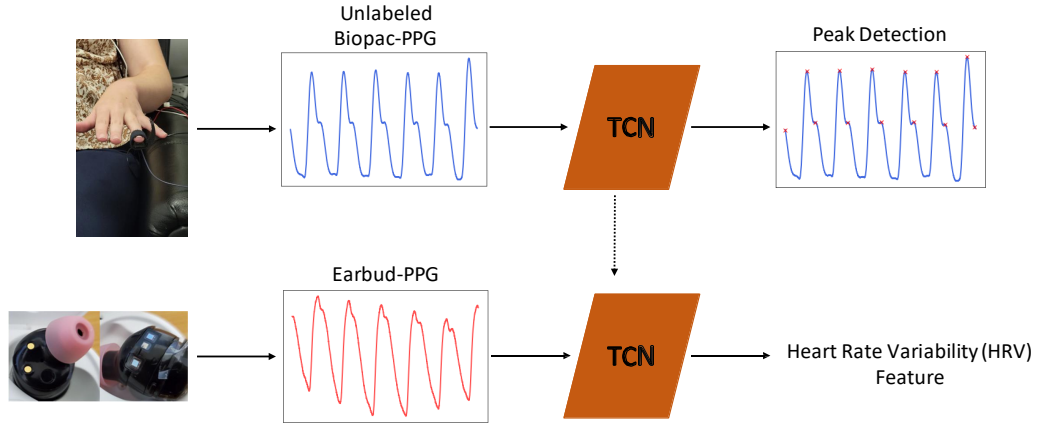


Fig. 2: Transfer learning framework to predict HRV features

instead of standard convolutions and using a receptive field that is exponential in network depth.

B. Downstream HRV parameters estimation

The pre-trained TCN model from the previous step is fine-tuned to predict different HRV-based features from the earbud PPG signal. The trained TCN network is transferred here, with only the final layer changed to predict various HRV parameters instead of peaks. In our current work, we considered five different features, including the mean heart rate (HR) and the following HRV features: root mean square of successive differences between normal heartbeats (RMSSD), proportion of the number of pairs of successive NN (R-R) intervals that differ by more than 50 ms (pNN50), standard deviation of normal to normal R-R intervals (SDNN), and normalized high frequency power (HFnorm). We fine-tuned the pre-trained model separately for each of these HRV parameters. As we considered features that are continuous values, this required changing only the loss function from the pre-trained model.

While our framework can be extended to predict any HRV-based parameter, we chose these five as they were the most predictive of stress in our prior empirical analysis [16]. An overview of our framework is shown in Figure 2.

IV. RESULTS

Using the framework described in the previous section, we initially trained the upstream model to predict the peaks of the input Biopac signal. This upstream model was subsequently fine-tuned on earbud PPG signal to predict different features separately: HR, RMSSD, pNN50, SDNN, and HFnorm. Ultimately, using these features, we predict whether the subject is feeling stress or not for each PPG signal window. We conducted all our analysis using Leave-One-Subject-Out Cross-Validation (LOSOXV), where the data from a single subject was held out for testing while the model was trained on the remaining subjects. This process was repeated until all subjects had been tested.

	HeartPy	Our framework	Our framework without pre-training
HRV feature	MAE	MAE	MAE
<i>HR</i>	5.9 ± 2.6	10.8 ± 3	10 ± 3.28
<i>RMSSD</i>	16.7 ± 2.4	11.2 ± 2.9	14.74 ± 7.9
<i>pNN50</i>	0.17 ± 0.03	0.14 ± 0.02	0.15 ± 0.03
<i>SDNN</i>	27.2 ± 8.1	17.3 ± 3.5	20.7 ± 7.9
<i>HFnorm</i>	28.4 ± 5.8	23.2 ± 2.6	24.5 ± 4.3

TABLE I: Comparison of HRV feature estimation

A. Feature Estimation

The feature values predicted by our model are compared against HeartPy predictions. To the best of our knowledge, HeartPy is the most prominent and trusted public library to provide HRV features from PPG signals. For the prediction of all the HRV features, we use a 4-layer TCN to get the input representation from the PPG signal that, in turn is passed through 2 Fully Connected (FC) layers with hidden sizes of 64 to get the output prediction. We train the model using a learning rate of 5e-05 and a dropout value of 0.3. We evaluate the performance of our model against HeartPy using mean absolute error (MAE) with the estimates from the Biopac sensor data considered the ground truth. We report these results in Table I. Among all the parameters predicted, we observe that for heart rate alone, HeartPy outperforms our proposed framework. For the remaining HRV parameters, our model performs better than HeartPy prediction. On average, the improvement in HRV parameter estimation compared to HeartPy is around 26%.

To highlight the significance of fine-tuning, we predicted HRV features from the earbud PPG without any pre-training step. The results of this experiment are shown in Table I. With the exception of HR, we observe that transfer learning improves performance for all the HRV features.

Throughout all our experiments, we consistently observed relatively poorer performance in average HR estimation using our proposed framework. This may be related to the fact that heart rate estimation is fundamentally different from HRV parameter estimation. While HRV estimation depends on the specific sequence of individual peaks, HR is more of an average of the distances between peaks. The framework was not optimized for this averaging task and the removal of outliers that would adversely affect the average. Nevertheless, heart rate is an important feature and integrating its estimation in a unified framework can be the focus of future work.

B. Stress detection

In this section, we report the performance of stress classification using HRV parameters obtained from HeartPy and our model, respectively. The PPG window from a subject is labeled as stress if it is sampled from the task that induces stress, which in our case are Dot-tracking, Speech-Preparation, Speech, and Exciting Music tasks (see Figure 1 for more details), PPG windows from the remaining tasks are labeled

HeartPy feature replaced	Accuracy	Sensitivity	Specificity
-	0.81	0.74	0.83
<i>RMSSD</i>	0.86	0.80	0.87
<i>pNN50</i>	0.86	0.80	0.88
<i>SDNN</i>	0.86	0.79	0.87
<i>HFnorm</i>	0.87	0.81	0.88

TABLE II: Stress detection results when one of the HeartPy HRV features is replaced with our model predictions

as no-stress. As a first step, we classify stress using all the HRV features that the HeartPy library outputs from a window of PPG signal, using a random forest classifier developed in our previous work. In the next step, we replace one of the HeartPy's RMSSD, pNN50, SDNN, HFnorm with our model predictions and repeat the stress classification task. The increase/decrease in the performance of stress detection by replacing HeartPy predictions with our model predictions gives us a better understanding of how to evaluate these parameters. The results of this experiment are shown in Table II. We also experimented with stress detection by using all the features as predicted by our framework. However, given the prominent role of HR in estimating stress, this model's performance was relatively lower compared to ones reported here. We evaluate the performance of stress detection using Accuracy, Specificity, and Sensitivity. From this table, we can see that using the HRV parameter predicted by our framework, instead of the HeartPy prediction, has resulted in at least a 5% improvement in all evaluation metrics for all HRV features. This demonstrates the impact of our proposed framework on feature extraction in stress classification.

V. CONCLUSION

Estimating HRV features using wearable sensors is critical for stress management solutions. In our current work, we take an initial step toward using earbud PPG signals to estimate HRV features, employing a framework based on transfer learning and TCN architecture. We have demonstrated that our framework estimates these features more accurately than the existing open-source library, HeartPy. Subsequently, we observed that using our model's predicted HRV features in place of HeartPy for stress detection results in improved performance. Encouraged by these initial results, we plan to conduct our study on a larger and more diverse set of subjects to corroborate our findings.

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